

Syllabus - Stars and Universe

Fall Semester, 2025

Course Title	Stars and Universe	Course Code		PE4203	Credit	3	Classroom	Changjo 5510
		Type		Elective	Hours	3		
Teacher	Name	Class no.	Office	Office hours	Contact		Google Classroom	
	PARK Sun Mie	1	Changjo #5504	Mon. 14:40-16:30 Tue. 13:40-14:30 15:40-16:30	smpark88@gmail.com 051-606-2241		cqmlrxvs	

1. Course Description

Course Objectives
Students will deal with the astronomical observations quantitatively, and learn basic contents of modern astronomy and astrophysics. Students will explore the universe through student-centered problem solving learning and project-based learning.
Contents
Astronomy is the oldest of the natural sciences, but it is also a cutting-edge science that discovers new facts through the theoretical study of astrophysics and modern astronomical instrument such as the space telescope. Astronomy is characterized by enormous changes of scales from the size of the smallest particle (quark) to the largest structure (universe). It also covers all the time from the beginning of the universe to its possible end. So, The topics of astronomy encompass both problems which everyone wonder and which are the most difficult to challenge. This course deals with basic contents of modern astronomy and astrophysics including the celestial mechanics, solar system, evolution of stars, galaxies, and evolution of universe.

2. Core Target Competencies

Subject Core Competencies					
Ability to understand and apply scientific knowledge	Scientific thinking	Problem-solving skills	Ability to explore and experiment	Creativity	Ability to collaborate and communicate

3. Text & References

- A. Textbook: Introductory Astronomy and Astrophysics, A. M. Zeilik, S. A. Gregory, 4th Edition, BOOKS/COLE, 1998
- B. References: 태양계 천문학, 이시우, 안병호, 서울대학교출판부, 1997
- An Introduction to Modern Astrophysics, Bradley W. Carroll and Dale A. Ostlie, Addison Wesley Longman, 1996

4. Class Method

- A. Student-centered problem solving learning
- B. Project based learning
- C. Presentation and discussion

5. Evaluation : The grade will be given according to total points.

A. Evaluation Criteria

Problem solving, Presentation, Homework	Term Project	Midterm Exam	Final Exam	Attitude and Attendance	Sum(%)
25	15	25	25	10	100

B. Evaluation Method

(1) Problem solving and Presentation (Group/Individual evaluation)

- You should solve the learning problems through group discussion and present them in every class.
- Class progress : Introducing learning topic → Addressing problems → Group discussion and learning → Group presentation and Q&A → Wrap-up
- Evaluation standards : ① They present logically, interestingly and confidently?
② During the task, their roles are well organized? ③ They try to study creatively?
④ There is any further investigation? ⑤ They make good use of figures and tables?

(2) Homework (Individual evaluation)

- In time you will get the score range of A to D, and it will be normally 1 week and the exact deadline will be announced in the classroom. If you submit your report after that but within 1 week, then you will get a score degraded by 10% each day. After 1 week, any report cannot be accepted and the score is automatically F. If you have any problem, please consult with the instructor in advance.

* **Plagiarism** : A meeting of faculty committee will be open for FAIL of the course.

(3) Term Project (Group/Individual evaluation)

- You should study a topic related to this class for a semester and present the results.
- It is necessary to select a **scientifically meaningful topic** beyond simply having fun.
- Evaluation standards : ① research content, ② presentation, and ③ report
- The research content is assessed through the ① plan, ② process, and ③ result.
- The creativity, scientific logic, and cooperation will be checked.
- The presentation follows the above evaluation criteria.
- The report is assessed based on the completeness (introduction, method, result, and discussion).

(4) Attitude (Individual evaluation)

- They participate actively, collaboratively, and interestingly in class?
- They think and ask a lot of questions?

(5) Attendance

- Unexcused absence makes the score degraded by 10% each time, except illness (with confirmation from school infirmary and contact to instructor in advance) or official cases.

6. Remarks

☞ You need to

- Actively, interestingly participate in class.
- Think and ask a lot of questions to me and yourself.

7. Criteria for Awarding Credits

☞ Students with excellent academic achievement are recommended for the Academic Excellence Award and the Principal's Award, and are recorded according to core competencies in the Subject Details. However, students with poor academic attitude or poor attendance are excluded.

- Attendance for at least three-quarters of the total class hours is required to earn credit.
- For Pass/Fail courses, if the instructor determines that a specific evaluation element is insufficient and requires a separate evaluation, a separate evaluation may be conducted.
- Calculation of the basis score for awarding credit to absentees : The standard score (Z) from the taken examination will be converted into the corresponding standard score of the untaken examination.
- If a basis score cannot be calculated, the grading method : Article 10, Paragraphs 2 and 3 of the "Academic Performance Management Regulations" will apply accordingly.

※ Refer to the "Academic Performance Management Regulations" for the percentage of credit points applied.

8. Weekly Plans

Week (period)	Topic and Contents	(Notes)
Week 01 8.18.~8.22.	Introduction Ch01. Celestial Mechanics and the Solar System I	
Week 02 8.25.~8.29.	Ch01. Celestial Mechanics and the Solar System II	HW #1
Week 03 9.1.~9.5.	Ch01. Celestial Mechanics and the Solar System III	HW #2 TP
Week 04 9.8.~9.12.	Ch01. Celestial Mechanics and the Solar System IV	HW #3 TP
Week 05 9.15.~9.19.	Ch08. Electromagnetic Radiation and Matter I	TP
Week 06 9.22.~9.26.	Ch08. Electromagnetic Radiation and Matter II	TP : Determination of topic for TP (Presentation and Submission of research proposal)
Week 07 9.29.~10.3.	Ch08. Electromagnetic Radiation and Matter III	HW #4
Week 08 10.6.~10.10.	Chuseok Holiday, Hangeul Day, Discretionary Holiday	
Week 09 10.13.~10.17.	Mid-term Examination (10.13.~10.16.)	10.17. Regular classes
Week 10 10.20.~10.24.	Ch13. Spectrum and The Hertzsprung-Russell Diagram	HW #5 TP
Week 11 10.27.~10.31.	Ch15. The Interstellar Medium and Star Birth	
Week 12 11.3.~11.7.	Ch16. The Evolution of Stars I	TP : Interim presentation
Week 13 11.10.~11.14.	Ch16. The Evolution of Stars II	HW #6
Week 14 11.17.~11.21.	Ch17. Star Deaths I	TP : Interim presentation
Week 15 11.24.~11.28.	Ch17. Star Deaths II	Wrap up TP
Week 16 12.1.~12.5.	Ch26. The New Cosmology	Make-up class week TP : Presentation of TP result
Week 17 12.8.~12.12.	Final Examination (12.08.~12.12.)	

※ The above plans may change according to the academic schedule.

9. Term Project Schedules

Week	Class	Contents	Remarks
1			
2			
3	1	Investigation of research field	
4	1	Investigation of research topic	
5	1	To specify and present research topic	ppt
6	2	Preparation and Submission of Research Proposal – Research overview – Research background and necessity (including previous studies) – Research objective – Research content and methods – Expected impacts of research results and potential application – Research timeline (weekly plan) – Reference * Submission of purchase application for necessary items	Determination of topic for TP
7			
8		Chuseok Holiday, Hangeul Day, Discretionary Holiday	
9		Mid-term Examinations	
10	1	Term Project	
11			
12	2	Term Project - Interim presentation	Evaluation of TP
13			
14	2	Term Project - Interim presentation	Evaluation of TP
15	2	Term Project – Wrap up	
16	2	Term Project – Presentation of result	ppt/poster
17		Final examinations	
Total	14		

※ The schedule will change as it may chance.

How to write an effective report

Almost IMRAD format:

Title - The title should reflect the factual content of the paper. Here you will write your name and the date on which the experiment was performed or the date the lab report was written.

Abstract – Summary of the basic content of the experiment. It should state the purpose of the experiment, mention the techniques used, report results obtained, and give conclusions. The point of the abstract is to give a concise summary of the whole report.

Introduction or Purpose – It should provide the reader with information necessary to understand the scientific basis of the experiment and the techniques used. In most cases, you should include background information and explain the theory behind the techniques. The most common mistake that students make is references. When you refer any resources from others, you should note the source of the information. You are not supposed to write introduction in the amount of half of the report. Keep in mind to write concisely key information and key technique.

Methods and Material – It should be included all raw data, observations, calculations and appropriate graphs.

Results and Analysis – A summary of the final data or output. A short paragraph should be sufficient, along with any relevant charts and graphs labelled well. Remember to title and provide legends for all graphs and tables. The graphs and tables should be comprehensible independently of their association with the text.

Discussion or Summary - Discuss the experiment and the results obtained. **Describe the key findings from the results. Interpret and discuss their significance.** Discuss whether your interpretations support or refute your hypothesis. **Discuss the reasons for your findings,** especially the main findings. Compare and contrast your findings with other studies. Present the strengths and weaknesses of your study including limitations. Suggest further research opportunities. **Make a final summary and conclusion.** Clearly state your overall findings and their implications.

Reference Citations - As required in all scientific literature, statements of fact, not considered "common" knowledge, must be properly referenced.

- Note 1: This kind of IMRAD format is a guideline which helps you to write a report. You don't have to follow this scheme, but you can rather re-organize the report depending on the topic if you want.
- Note 2: This guideline is adapted from one in the Writing Center, the University of Wisconsin and modified for this class suitably.

References:

References must be obviously related to the manuscript.

In the text, references should be cited with the author's surname and year of publication (e.g., Kim, 2000). When reference is made to a work by two authors, both names should be given using "&" (e.g., Kim & Lee, 1996); for three or more author names, give the first author followed by "et al." and the year (e.g., Park et al., 2010). Multiple references must be arranged in chronological order (e.g., Ahn et al. 2003; Choi & Kang 1991; Sohn 1982). If more than two papers with the same authors and publication years are cited, list a, b, c... after the year to clarify (e.g., Jung & Han 2011a, b). Only cite articles or books already published or in press, not unpublished work "in preparation".

In the references section, the references should be listed in alphabetical order by letter of the first author's surname. List the first five authors followed by "et al." if there are more than five authors. If more than two papers with the same authors and publication years are cited, list a, b, c... after the year in publication order. Abstracts of conferences should not be included in the references. Lines after the first line of a reference are indented by one tab space ("hanging indent" in MS Word). The style for citing papers in periodicals is surname and initials of authors, title, journal name, volume, first and last page numbers, year, and digital object identifier (DOI) if available. The styles to be used for references are as follows:

Journal articles:

1. Takahashi K., McPherron R.L., Hughes W.J., Multispacecraft observations of the harmonic structure of Pc3-4 magnetic pulsations, *J. Geophys. Res.* 89, 6758-6774 (1984).
<https://doi.org/10.1029/JA089iA08p06758>

Articles from book, conference or symposium proceedings:

1. Kelley M.C., The Earth's Ionosphere (Academic Press, San Diego, 1989), 123-157.
2. Capitaine N., Gambis D., McCarthy D.D., Petit G., Pay J., et al., Proceedings of the IERS Workshop on the Implementation of the New IAU Resolutions, Observatoire de Paris, Paris, France, 18-19 April 2002.
3. Kim K., Hwang J., Sung S., Geosynchronous magnetic field variations associated with the passage of interplanetary shocks or solar wind discontinuities, in 2007 AGU Fall Meeting, San Francisco, CA, 10-14 Dec 2007.

Technical report:

1. Mazanek D.D., Roithmayr C.M., Antol J., Park S.Y., Koons R.H., et al., Comet/asteroid protection system (CAPS): preliminary space-based system concept and study results, NASA Langley Research Center Technical Report, NASA/TM-2005-213758 (2005). ₩

Dissertation:

1. TPark S.Y., Optimization and guidance of ascent trajectories with inequality constraints, PhD Dissertation, Texas A&M University (1996).

Website:

1. TNational Radio Astronomy Observatory, Very long baseline array (2009) [Internet], viewed 2011 Feb 20, available from: <https://www.vlba.nrao.edu>

- Note 1: This guideline is adapted from the "Instruction for Authors" in the Journal of Astronomy and Space Science and modified for this class suitably

Rubric for Lab Report

Student' Name:

Credit:

1. Abstract
 - A. Abstract contains reference to all major aspects of carrying out the experiment and the results, well-written
 - B. Abstract references most of the major aspects of the experiment, some minor details are missing
 - C. Abstract misses one or more major aspects of carrying out the experiment or the results
2. Experimental Procedure
 - A. Well-written in paragraph format, all experimental details are covered
 - B. Written in paragraph format, important experimental details are covered, some minor details missing
 - C. Written in paragraph format, still missing some important experimental details
3. Results: data, figures, graphs, tables, etc.
 - A. All figures, graphs, tables are correctly drawn, are numbered and contain titles/captions
 - B. All figures, graphs, tables are correctly drawn, but some have minor problems or could still be improved
 - C. Most figures, graphs, tables OK, some still missing some important or required features
4. Discussion
 - A. Important all knowledge and data comparisons have been interpreted correctly and discussed, good understanding of results is conveyed
 - B. Almost all of the results have been correctly interpreted and discussed, only minor improvements are needed
 - C. Some of the results have been correctly interpreted and discussed; partial but incomplete understanding of results is still evident
5. Spelling, grammar, sentence structure
 - A. All grammar/spelling correct and very well-written
 - B. Less than 3 grammar/spelling errors, mature, readable style
 - C. Occasional grammar/spelling errors, generally readable with some rough spots in writing style
6. Appearance and formatting
 - A. All sections in order, well-formatted, very readable
 - B. All sections in order, formatting generally good but could still be improved
 - C. Sections in order, contains the minimum allowable amount of handwritten copy, formatting is rough but readable

- NOTE: This rubric is adapted from one in the Winona State University and is modified for this class suitably